

6. (a) The table shows some information about substances **A-E**.

Substance	Soluble in water?	Malleable?	Boiling point (°C)	Conducts electricity?
A	no	no	4 200	no
B	yes	no	-79	no
C	yes	no	1 413	only when molten or in solution
D	no	yes	5 555	yes
E	no	yes	2 562	yes

(i) Give the **letters** of the **two** substances which are metals.

[1]

..... and

(ii) Give the **letter** of the substance which could be carbon dioxide.

[1]

.....

(iii) Give the **letter** of the substance which has a giant ionic structure.

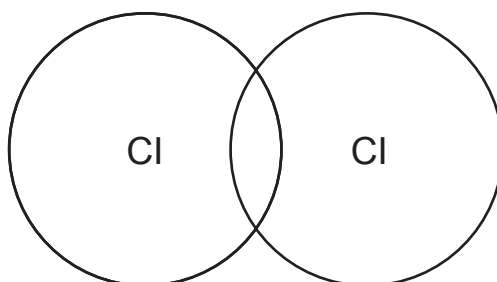
[1]

.....

(b) Complete the dot and cross diagram to show the outer electrons in a chlorine molecule, Cl_2 .

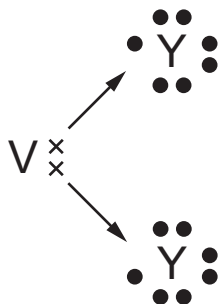
Each atom of chlorine has 7 electrons in its outer shell.

[2]



- (c) The diagram below shows the electronic changes that occur when atoms **V** and **Y** form ionic bonds. The dots and crosses show outer shell electrons.

V and **Y** are **not** the chemical symbols of the elements.



- (i) Give the **charge** on each ion produced. [2]

V

Y

- (ii) Circle the formula of the compound formed. [1]



- (iii) Potassium oxide, K_2O , is another ionic compound.

The relative formula mass of potassium oxide is 94.

Calculate the percentage of potassium in potassium oxide, K_2O . Give your answer to the nearest **whole number**. [2]

$$A_r(\text{K}) = 39$$

Percentage = %

